

Validation Execution Completeness & Conformity Module (VECCM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
Parent Standard: Validation Execution Standard (VEXS)
Operational Layer: Validation Execution Governance Layer
Category: AI & Interpretation
Subcategory: Validation Governance
Type: Parent Standard Module
Governed Space: Validation Execution Completeness & Conformity
Version: 1.0
Status: Canonical · Open Module
Effective Date: 11 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Execution Completeness & Conformity

Canonical Definition

Validation Execution Completeness & Conformity Module (VECCM)

defines the governance framework for determining whether a Validation Execution Instance has performed the materially required validation activities with sufficient completeness, conformity, continuity, control, reconstructability, and resolution of material deviations to support progression into formal Validation Determination.

It establishes the final execution-governance gate through which the **Actual Validation Execution Record** is compared against the **Authorized Validation Execution Basis** and classified according to its methodological fitness for subsequent determination.

Operational Role

VECCM serves as the fifth and final internal governance space of the Validation Execution Standard.

The preceding modules establish:

VEBAM — Validation Execution Basis & Authorization

VEICM — Validation Execution Instantiation & Control

VEOTM — Validation Execution Observation & Traceability

VEDEGM — Validation Execution Deviation & Exception Governance

VECCM asks:

Considering what was authorized, what was actually performed, what can be reconstructed, and which deviations remain, is the complete validation execution sufficiently complete and conformant to support a legitimate Validation Determination?

The governed progression is:

Authorized Execution Basis → Actual Execution Record → Completeness Assessment → Conformity Assessment → Residual Deviation Assessment → Reconstruction Assessment → Determination Readiness → Execution Conformity Determination

VECCM does not determine whether the Validation Object satisfies the Validation Criteria.

It determines whether the validation execution itself is methodologically usable for that later determination.

Module Operational Space

VECCM governs:

- execution completeness,
- execution conformity,
- execution coverage,
- required-activity completion,
- criteria-execution completeness,
- method-step completeness,
- execution continuity,
- object continuity,
- scope conformity,
- method conformity,
- evidence-use conformity,
- source-condition conformity,
- configuration conformity,
- control conformity,
- sequence conformity,
- participant-role conformity,
- reconstruction sufficiency,
- traceability sufficiency,
- residual deviation impact,
- unresolved exception impact,
- execution integrity readiness,
- conditional conformity,
- execution invalidity,

- Determination Readiness,
 - Determination Block,
 - execution completion status,
 - execution conformity status,
 - execution reassessment,
 - and execution-conformity traceability.
-

Module Function

VECCM applies when material validation execution activities have ended, been terminated, or reached a stage where progression to Validation Determination is being considered.

Its function is to prevent:

- partially executed validation being represented as complete,
 - complete activity counts being mistaken for methodological conformity,
 - unresolved deviations disappearing at execution closure,
 - missing criteria being hidden by successful execution elsewhere,
 - scope reduction being forgotten,
 - wrong object versions being overlooked,
 - incomplete traceability being ignored,
 - source restrictions being lost,
 - execution under changed configurations being treated as equivalent,
 - failed or omitted activities being silently excluded,
 - and a final Validation Determination being issued where the execution record cannot legitimately support one.
-

Completeness and Conformity Are Different

VALIDOS® establishes a strict distinction:

Execution Completeness asks:

Were the materially required validation activities actually performed?

Execution Conformity asks:

Were those activities performed with sufficient correspondence to the Authorized Validation Execution Basis?

A validation execution may therefore be:

Complete but Non-Conformant

or:

Conformant in performed activities but Incomplete overall

For example:

Every executed test may follow the method perfectly, while one mandatory test was never performed.

That execution is not complete.

Likewise, every required activity may have occurred, but several were conducted against the wrong configuration.

That execution may be complete but materially non-conformant.

Execution Completeness

Execution Completeness evaluates whether all materially required validation activities have been addressed.

This may include completeness across:

- Validation Scope,
- Validation Criteria,
- method steps,
- required evidence interactions,
- required source interactions,
- required operating conditions,
- required scenarios,
- required repetitions,
- required observations,
- required reviews,
- and required controls.

Completeness should remain proportionate to the approved Validation Purpose and Validation Depth.

Required Activity Completion

A required activity may be classified as:

Completed

Conditionally Completed

Partially Completed

Not Completed

Not Applicable

or:

Completion Undetermined

These classifications should remain traceable to the

execution basis.

Criteria Execution Completeness

Every material Validation Criterion requiring execution should have sufficient execution activity associated with it.

VECCM asks:

- Was the criterion actually examined?
- Were all required activities performed?
- Were required conditions represented?
- Was execution sufficient for later criterion determination?
- Are material parts of the criterion still unsupported?

This does not determine whether the criterion was satisfied.

Method-Step Completeness

Where the Validation Method requires a defined sequence or set of steps, VECCM determines whether materially required steps were completed.

Missing non-critical procedural detail may not always block determination.

Missing a methodologically essential step may.

Materiality must therefore remain explicit.

Scope Completeness

Validation execution must be compared against the approved Validation Scope.

Possible conditions include:

Scope Fully Executed

Scope Conditionally Executed

Scope Partially Executed

Scope Materially Reduced

or:

Scope Execution Undetermined

Where actual execution covers less than the authorized scope, later Validation Determination must remain bounded accordingly.

Validation Depth Completeness

VECCM assesses whether execution achieved the required Validation Depth.

A validation may perform all nominal activities while failing to achieve the depth required for:

- high-consequence reliance,
- enhanced assurance,
- regulatory use,
- critical deployment,
- or another governed purpose.

Therefore:

Activity Completion ≠ Required Validation Depth Achieved

Execution Conformity

Execution Conformity evaluates the degree to which Actual Validation Execution corresponds to the Authorized Validation Execution Basis.

Conformity may include:

- correct Validation Object,
- correct object version,
- approved scope,
- approved method,
- approved configuration,
- required sequence,
- required evidence,
- governed sources,
- required controls,
- authorized participants,
- required environment,
- and applicable restrictions.

Object Conformity

VECCM verifies whether execution remained sufficiently connected to the Validation Object authorized for validation.

Relevant questions include:

- Was the correct object examined?
- Was the correct version used?
- Did the object materially change during execution?
- Were material state changes governed?
- Did outputs remain attributable to the correct object state?

A critical object mismatch may invalidate affected execution.

Scope Conformity

Execution may move outside or inside the authorized scope.

VECCM determines whether:

- required scope was preserved,
 - unauthorized scope expansion occurred,
 - material scope reduction occurred,
 - exclusions were governed,
 - and actual scope can be reliably defined.
-

Method Conformity

VECCM evaluates whether the instantiated and performed method remained sufficiently aligned with the approved Validation Method.

This includes consideration of:

- required method steps,
 - parameters,
 - sequence,
 - repetition,
 - comparison structure,
 - analytical conditions,
 - simulation conditions,
 - or another method-specific requirement.
-

Configuration Conformity

Validation may depend materially upon:

- model configuration,
- system settings,
- instrument settings,
- environment settings,
- software versions,
- dataset versions,
- or simulation configurations.

VECCM determines whether actual configurations remained sufficiently aligned with those authorized.

Evidence-Use Conformity

VECCM assesses whether validation used evidence in accordance with its approved role and Evidence Fitness conditions.

This may include:

- correct evidence identity,
- evidence version,
- permitted role,
- applicability conditions,
- temporal validity,
- scope limitations,
- and any conditional fitness requirements.

Source-Condition Conformity

Where sources were subject to Source Reliability conditions, VECCM determines whether execution respected those conditions.

For example:

A source approved only with independent corroboration should not be treated as unrestricted if the required corroboration was never performed.

Control Conformity

Required controls may include:

- access controls,
- logging,
- environmental control,
- sequence control,
- oversight,
- configuration locks,
- safety controls,
- or validation-specific controls.

VECCM determines whether required controls were sufficiently active across the activities they were intended to govern.

Sequence Conformity

Where sequence matters methodologically, VECCM assesses whether activities occurred in the required order.

Sequence deviation may affect only one activity or the validity of a larger methodological chain.

Participant-Role Conformity

VECCM may assess whether material validation activities were performed by participants or systems possessing the required execution role.

This may include:

- authorization,
- role boundaries,
- competence conditions,
- independence requirements,
- or permitted intervention rights

where those conditions formed part of the Execution Basis.

Execution Continuity

Execution Continuity concerns whether the validation process remained sufficiently connected across its material stages.

Continuity may be affected by:

- long interruptions,
- object changes,
- evidence replacement,
- source change,
- environment change,
- configuration change,
- participant change,
- or execution restart.

A resumed execution should not automatically be treated as continuous where material conditions changed.

Traceability Sufficiency

VECCM consumes VEOTM outputs to determine whether the execution history is sufficiently traceable for conformity assessment.

A validation may have been performed correctly but lack sufficient records to demonstrate that fact.

VALIDOS® establishes:

Correct Execution That Cannot Be Demonstrated ≠ Fully Governed Execution

where demonstrability is required for the intended validation role.

Reconstruction Sufficiency

VECCM asks whether a qualified reviewer or system can sufficiently reconstruct:

- what occurred,
- against which object,
- using which method,
- with which evidence,
- from which sources,
- under which conditions,
- in what sequence,
- with which deviations,
- and with which outputs.

Reconstruction need not be perfect.

It must be sufficient for the intended Validation Determination.

Residual Deviation Assessment

VECCM consumes the complete deviation and exception record established under VEDEGM.

It determines what residual effects remain after:

- correction,
- compensation,
- re-execution,
- escalation,
- or closure.

A closed deviation may still leave a material methodological limitation.

Resolved Does Not Mean Immaterial

VALIDOS® establishes:

Deviation Resolved ≠ Deviation Never Mattered

A resolved deviation may continue to affect:

- scope,
 - interpretation,
 - confidence,
 - evidence use,
 - or later reliance.
-

Residual impact must remain visible.

Unresolved Deviation

A material unresolved deviation may prevent unrestricted Execution Conformity.

Depending upon materiality, it may lead to:

- Conditional Conformity,
 - Non-Conformity,
 - Determination Block,
 - or Execution Invalidity.
-

Cumulative Deviation Assessment

VECCM evaluates whether multiple minor or moderate deviations collectively create a material conformity problem.

This prevents fragmented exception handling from missing system-level execution degradation.

Execution Completeness Status

VECCM may establish:

Execution Complete

All materially required validation activities have been sufficiently performed.

Execution Conditionally Complete

Execution is complete only within explicit boundaries, limitations, or reduced scope.

Execution Partially Complete

Material validation activities remain unperformed.

Execution Incomplete

The execution cannot reasonably be treated as complete for the intended Validation Purpose.

Execution Completeness Undetermined

Available records are insufficient to establish completeness.

Execution Conformity Status

VECCM may establish:

Execution Conformant

Actual validation execution sufficiently corresponds to the Authorized Validation Execution Basis for the intended determination.

Execution Conditionally Conformant

Execution may support Validation Determination only under explicit conditions, limitations, scope restrictions, or interpretive boundaries.

Execution Conformity Undetermined

Available information is insufficient to establish whether execution was sufficiently conformant.

Execution Non-Conformant

Material execution conditions departed from the authorized basis sufficiently to prevent unrestricted methodological reliance.

Execution Invalid

Critical execution deficiencies prevent the affected execution from legitimately supporting the intended Validation Determination.

These states describe execution.

They do not describe the Validation Object.

Conditional Conformity

Conditional Conformity may arise where:

- one validation condition remains limited,
- scope was reduced,
- one source required restriction,
- a deviation was compensated but leaves residual uncertainty,
- a subset of criteria remains usable,
- or validation can support only a narrower determination than originally intended.

Conditions must remain attached to the subsequent Validation Determination.

Partial Execution Validity

Where execution is partly valid and partly invalid, VALIDOS® should avoid unnecessary all-or-nothing treatment where methodological separation is possible.

VECCM may identify:

Valid Execution Portion

and:

Invalid or Unusable Execution Portion

provided the boundaries can be reliably established.

This supports precise remediation and partial re-execution.

Execution Invalidity

Execution Invalidity should be reserved for conditions that fundamentally prevent legitimate methodological use of affected execution.

Potential causes include:

- wrong Validation Object,
 - critical object-version mismatch,
 - critical method failure,
 - irrecoverable sequence corruption,
 - critical evidence discontinuity,
 - critical source-condition failure,
 - untraceable critical activity,
 - severe uncontrolled deviation,
 - or inability to reconstruct the execution sufficiently.
-

Invalid Execution Does Not Mean Invalid Object

VALIDOS® establishes one of its strongest distinctions:

Execution Invalid ≠ Validation Object Invalid

Invalid execution means the validation process cannot legitimately establish the intended conclusion.

The Validation Object may still satisfy or fail the Validation Criteria.

VALIDOS® simply does not claim to know on the basis of that execution.

Conformant Execution Can Demonstrate Failure

Conversely:

Execution Conformant + Criterion Failure = Legitimate Negative Validation Finding

A validation process should not be judged by whether it produces a positive result.

It should be judged by whether it produces a legitimate result.

Determination Readiness

VECCM creates the final execution gate into Validation Determination.

A Validation Execution Instance may be considered **Determination Ready** where sufficient basis exists to determine:

- what was executed,
 - which criteria were examined,
 - which execution outputs exist,
 - how complete execution was,
 - how conformant execution was,
 - which deviations occurred,
 - what limitations remain,
 - and which portions of the execution may legitimately support determination.
-

Determination Readiness Status

Possible statuses may include: **Determination Ready**

Execution is sufficiently complete, conformant, reconstructable, and governed to support Validation Determination.

Determination Conditionally Ready

Determination may proceed only within explicit conditions, restrictions, limitations, or reduced scope.

Determination Readiness Undetermined

Available execution information is insufficient to determine readiness.

Determination Not Ready

Material execution deficiencies must be resolved before legitimate determination can occur.

Determination Blocked

One or more critical execution conditions prevent Validation Determination from legitimately proceeding.

Determination Block

A **Determination Block** prevents the architecture from converting deficient execution into a validity conclusion.

Potential triggers include:

- critical missing validation activity,
- wrong Validation Object,
- critical scope failure,
- execution invalidity,
- unreconstructable critical activity,
- unresolved critical deviation,
- critical method substitution,
- or another execution condition destroying the methodological basis for determination.

Determination Block Is Not Object Failure

A Determination Block means:

VALIDOS® cannot legitimately determine the object's validation result from this execution.

It does not mean:

The object failed validation.

This preserves methodological neutrality.

Execution Completion Record

VECCM may establish a formal Validation Execution Completion Record containing:

- Execution Instance,
- Validation Object reference,
- actual scope,
- actual Validation Depth,
- executed criteria,
- method execution summary,
- evidence used,
- sources used,
- execution outputs,
- deviations,
- exceptions,
- residual limitations,
- Execution Completeness Status,
- Execution Conformity Status,
- reconstruction status,
- Determination Readiness,
- and any Determination Block.

Execution Closure

Execution may be formally closed when:

- active execution has ended,
- material outputs are preserved,
- material deviations have status,
- completeness is assessed,
- conformity is assessed,
- residual limitations are recorded,
- and Determination Readiness has been established.

Closure does not erase the possibility of later reassessment.

Execution Reassessment

Execution completeness or conformity may require reassessment if:

- new execution records emerge,
 - a previously unknown deviation is discovered,
 - a source is later found unreliable,
 - evidence fitness changes,
 - a configuration discrepancy is discovered,
 - a traceability error is corrected,
 - or another material fact changes the execution picture.
-

Execution Conformity Change

A prior Conformant determination may become:

Conditionally Conformant

Non-Conformant

or:

Invalid

if new material information emerges.

Likewise, an initially uncertain execution may later become conformant after missing records or corrective execution become available.

Historical Execution Preservation

Reassessment should not erase earlier conformity determinations.

The record should preserve: **Original Determination** → **New Information**

→ Reassessment → Revised Determination

This creates methodological continuity and auditability.

Execution Handoff to Validation Determination

VECCM closes the Validation Execution lifecycle and passes forward a structured execution basis containing:

- actual performed activities,
- execution outputs,
- execution limitations,
- conformity conditions,
- residual deviations,
- scope boundaries,
- completeness status,
- and Determination Readiness.

The next Parent Standard can then ask:

What does this governed validation record actually demonstrate about the Validation Object?

Minimum Implementation Framework

1. Establish Actual Execution Scope

Determine what portion of the authorized validation scope was actually executed.

2. Assess Required Activity Completion

Determine whether materially required validation activities and method steps were completed.

3. Assess Criteria Execution Completeness

Determine whether each applicable Validation Criterion possesses sufficient execution activity for later determination.

4. Assess Execution Conformity

Compare Actual Validation Execution against the Authorized Validation Execution Basis across:

- object,
- scope,
- method,
- configuration,
- evidence,
- sources,
- sequence,

- controls,
- environment,
- and participant roles.

5. Assess Reconstruction & Traceability

Determine whether material execution history can be sufficiently reconstructed.

6. Integrate Deviations & Exceptions

Assess unresolved, residual, cumulative, and critical execution deviations.

7. Establish Completeness Status Classify execution completeness.

8. Establish Conformity Status

Classify execution conformity.

9. Establish Determination Readiness

Determine whether Validation Determination may:

- proceed,
- proceed conditionally,
- remain pending,
- or be blocked.

10. Preserve Lifecycle Traceability

Maintain:

- Execution Instance,
- Authorized Execution Basis,
- Actual Execution Scope,
- executed criteria,
- completed method steps,
- evidence used,
- sources used,
- execution outputs,
- traceability assessment,
- reconstruction assessment,
- deviations,
- exceptions,
- residual limitations,
- Execution Completeness Status,
- Execution Conformity Status,
- Determination Readiness,
- Determination Blocks,
- reassessments,
- versions,
- and sufficient lifecycle continuity.

Governance Outputs

VECCM may produce:

- Validation Execution Completeness Assessment,
- Required Activity Completion Map,
- Validation Criteria Execution Completeness Map,
- Method-Step Completion Record,
- Actual Validation Scope Record,
- Validation Depth Execution Assessment,
- Object Conformity Assessment,
- Scope Conformity Assessment,
- Method Conformity Assessment,
- Configuration Conformity Assessment,
- Evidence-Use Conformity Assessment,
- Source-Condition Conformity Assessment,
- Control Conformity Assessment,
- Sequence Conformity Assessment,
- Participant-Role Conformity Assessment,
- Execution Continuity Assessment,
- Traceability Sufficiency Assessment,
- Reconstruction Sufficiency Assessment,
- Residual Deviation Assessment,
- Cumulative Deviation Assessment,
- Execution Completeness Status,
- Execution Conformity Status,
- Valid Execution Portion Record,
- Invalid Execution Portion Record,
- Validation Execution Completion Record,
- Determination Readiness Status,
- Determination Block,
- Execution Closure Record,
- Execution Reassessment Record,
- and Execution Conformity Change Record.

Use Case 1 — AI Model Validation

Scenario

An AI model completes 98 of 100 planned validation activities.

The two omitted tests concern a critical autonomous-escalation criterion.

All other activities were performed correctly.

Application

VECCM does not classify the execution as complete merely because 98% of activities were performed.

The missing activities are evaluated according to their

methodological significance.

Result

Execution may be:

Partially Complete

and:

Determination Not Ready

for the affected criterion despite otherwise strong execution conformity.

Use Case 2 — Industrial System Validation

Scenario

All required industrial validation tests are completed, but one section of testing was performed after an unauthorized configuration change.

Application

VECCM distinguishes completeness from conformity.

Result

Execution may be:

Execution Complete

but: **Execution Conditionally Conformant**

or:

Execution Non-Conformant

for the affected section, depending upon materiality.

The relevant portion may require re-execution without discarding unaffected activities.

Use Case 3 — Autonomous Validation Agent

Scenario

An autonomous validation agent completes every required test and generates extensive records.

Later review discovers that several high-risk tests used an unapproved tool outside the authorized execution configuration.

Application

VECCM integrates:

- complete activity coverage,
- traceability,
- the tool-related deviation,
- affected criteria,
- and residual methodological impact.

Result

The validation cannot derive unrestricted confidence merely from complete automation.

The affected execution portion is classified according to conformity rather than activity count.

Architectural Position

Validation Execution Completeness & Conformity is the fifth and final internal governance space of the **Validation Execution Standard (VEXS)**.

The complete VEXS progression is:

Validation Execution Basis & Authorization → Validation Execution Instantiation & Control → Validation Execution Observation & Traceability → Validation Execution Deviation & Exception Governance → Validation Execution Completeness & Conformity

Together, these five governed spaces transform an approved validation design into a demonstrable and methodologically classified Validation Execution Record.

The lifecycle progresses:

Authorized Validation Basis

↓

Controlled Execution Instance

↓

Observed & Reconstructable Execution

↓ Governed Deviations & Exceptions

↓

Execution Completeness & Conformity



Determination Readiness

The next VALIDOS® Parent Standard can therefore address a fundamentally different question:

What does the complete governed validation record actually demonstrate about the Validation Object?

Foundational Principle

Validation execution becomes usable for determination only when what was required, what was performed, and what can be demonstrated are sufficiently aligned.

Canonical Closing Statement

Validation Execution Completeness & Conformity Module completes the execution-governance lifecycle within VALIDOS® by determining whether actual validation activity is sufficiently complete, conformant, reconstructable, controlled, and methodologically usable for subsequent Validation Determination. By governing required-activity completion, criteria execution, scope and depth, object and method conformity, evidence and source conditions, configurations, controls, sequencing, traceability, residual deviations, conditional conformity, execution invalidity, Determination Readiness, Determination Blocks, reassessment, and lifecycle continuity, VECCM ensures that VALIDOS® advances from execution to determination only when the operational validation record can legitimately support the conclusion it is being asked to produce.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Execution Deviation & Exception Governance Module \(VEDEGM\)](#)

Related Documents

→ [Validation Execution Standard](#)

→ [About Validation Execution Standard](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>